# Blessed G/Michael Catholic Secondary School

## Semester 1 Biology Mid1 Examination for Grade9

Academic Year: 2024 / 2018 E.C. Time Allowed: 1:30'

**Name:** _____ **Grade & Sec:** Grade9 _____ **R.No:** _____

**I. Write 'True' if the statement is correct and 'False' if it is incorrect.**

____ **1.** Dichotomous keys are useful for taxonomists to classify unfamiliar organisms.

____ **2.** Biology is related to chemistry and physics during studying the process of photosynthesis.

____ **3.** The correct way of handling/carrying a microscope is holding it with one hand only.

____ **4.** It is permissible to taste and sniff chemicals in laboratory rooms.

____ **5.** In symbiotic association between fungi and termites, fungi supply extracellular hydrolytic enzymes to break down cellulose.

**II. Match the items under Column A with the correct descriptions under Column B.**

**30.** Which pair correctly matches the organism with its cell wall chemical composition?

**A.** Cyanobacteria — Peptidoglycan cell wall
**B.** Cyanophyta — Cellulose cell wall
**C.** Yeast — Chitin & glucan cell wall
**D.** All of the above are correct

**III. Choose the best answer among the given alternatives.**

____ **6.** Which part of the optical microscope is used to move the mechanical stage rapidly for initial focusing?

**A.** Fine adjustment knob **B.** Coarse adjustment knob
**C.** Condenser **D.** Diaphragm

____ **7.** Which part of the microscope moves the stage slowly to produce a sharp, crisp focus?

**A.** Fine adjustment knob **B.** Coarse adjustment knob
**C.** Ocular lens **D.** Stage clips

____ **8.** Which component of a microscope controls the intensity and amount of light reaching the specimen?

**A.** Diaphragm **B.** Condenser **C.** Coarse knob **D.** Nosepiece

____ **9.** Which microscope part collects and focuses light rays from the illuminator onto the specimen?

**A.** Condenser **B.** Diaphragm **C.** Objective lens **D.** Mirror

____ **10.** Which apparatus is used in laboratories to provide high-pressure steam for complete sterilization?

**A.** Autoclave **B.** Pressure cooker
**C.** Incubator **D.** Hot oven

____ **11.** The ancient Greek philosopher who divided animals into two groups (those with red blood and those without) was:

**A.** Whittaker **B.** T.O. Diener **C.** Aristotle **D.** W.M. Stanley

____ **12.** The criteria used in R.H. Whittaker's five-kingdom classification system include:

**A.** Cell structure (prokaryotic vs. eukaryotic)
**B.** Mode of nutrition and reproduction
**C.** Phylogenetic evolutionary relationships
**D.** All of these

____ **13.** Which of the following biological groups are included within Whittaker's five kingdoms?

**A.** Viruses **B.** Bacteria
**C.** Lichens **D.** Bacteria and Algae

____ **14.** The naturalist who developed binomial nomenclature and authored "Systema Naturae" is:

**A.** Carolus Linnaeus **B.** Aristotle
**C.** Charles Darwin **D.** Jean-Baptiste Lamarck

____ **15.** What primary method of asexual reproduction is used by yeast?

**A.** Sexual spore formation **B.** Budding

**C.** Regeneration **D.** Conjugation

____ **16.** Protists are ecologically vital because they:

**A.** Produce oxygen via phytoplankton photosynthesis
**B.** Support marine and freshwater food chains
**C.** Recycle organic nutrients in aquatic habitats
**D.** All of the above

____ **17.** Fungi are indispensable to global ecosystems primarily because they function as:

**A.** Producers **B.** Regulators

**C.** Decomposers (saprotrophs) **D.** Controllers

____ **18.** Which statement is true regarding a well-designed controlled experiment?

**A.** Demonstrates cause-and-effect relationships
**B.** Isolates one independent variable while measuring a dependent variable
**C.** Keeps all other environmental variables strictly controlled
**D.** All are true expressions

____ **19.** Which of the following is an example of a non-hyphal (unicellular) fungus?

**A.** Yeast **B.** Mucor **C.** Rhizopus **D.** Mushroom

____ **20.** Which of the following cellular structures are used as locomotive appendages in unicellular organisms?

**A.** Flagella **B.** Cilia

**C.** Pseudopodia **D.** All of the above

____ **21.** Which of the following protists is free-living and does NOT cause human infectious disease?

**A.** Amoeba (Entamoeba) **B.** Paramecium

**C.** Plasmodium **D.** Giardia

____ **22.** Which method of asexual reproduction is characteristic of Paramecium?

**A.** Budding **B.** Sporulation

**C.** Binary fission **D.** Multiple fission

____ **23.** Which of the following is NOT a difference between fungi and animals?

**A.** Fungi have chitinous cell walls whereas animals lack cell walls
**B.** Fungi are heterotrophs while animals are autotrophs
**C.** Fungi are absorptive heterotrophs while animals are ingestive heterotrophs
**D.** Fungi are non-motile while animals are motile

____ **24.** Fungi resemble animals because they are:

**A.** Heterotrophs (store glycogen, non-photosynthetic)
**B.** Heterosporous
**C.** Chemoautotrophs
**D.** Photoautotrophs

____ **25.** All of the following are asexual modes of reproduction in fungi EXCEPT:

**A.** Budding **B.** Sporulation **C.** Fragmentation **D.** Conjugation

____ **26.** The vegetative body of multicellular fungi (mycelium) is composed of microscopic tubular branched filaments called:

**A.** Hyphae **B.** Sporangium **C.** Rhizoids **D.** Spores

____ **27.** Protozoan protists are traditionally classified into classes primarily according to:

**A.** How they obtain nutrients
**B.** How they grow
**C.** Their mechanism of locomotion (cilia, flagella, pseudopodia)
**D.** How they reproduce

____ **28.** Which of the following Archaebacteria can thrive in aerobic, hypersaline environments?

**A.** Halophiles **B.** Methanogens

**C.** Thermoacidophiles **D.** Cyanobacteria

___ **29.** During binary fission, which statement accurately describes the resulting daughter cells?

**A.** They have half as many chromosomes as the parent cell
**B.** They have double the chromosome number
**C.** They have the same chromosome count as the parent but initial smaller cell volume
**D.** They contain no genetic material